Comprehensive Explainable AI (XAI) Report: Interpreting Breast Cancer Diagnostics via SHAP and LIME

Executive Summary

As machine learning classifiers achieve state-of-the-art predictive performance across critical domains, their internal decision-making mechanisms have grown increasingly opaque. In healthcare—specifically oncology—high empirical accuracy alone is insufficient for clinical adoption. Clinicians require transparent, verifiable, and interpretable reasoning to trust automated diagnostic tools.

This technical report provides a rigorous evaluation of two foundational Explainable AI (XAI) frameworks: **SHAP (SHapley Additive exPlanations)** and **LIME (Local Interpretable Model-agnostic Explanations)**. Both methods are applied to interpret predictions generated by an ensemble **Random Forest Classifier** trained on the Breast Cancer Wisconsin (Diagnostic) dataset. The trained model achieved an empirical test accuracy of **96.49%**. Global feature attribution via SHAP identified key morphological indicators—including worst concave points, worst perimeter, worst radius, and worst area—as the primary drivers of tumor malignancy classification. Local instance analysis for sample test index 0 demonstrated how SHAP and LIME converge on core decision drivers while exhibiting distinct theoretical behaviors resulting from exact game-theoretic attributions versus local linear surrogate approximations. Finally, strategic recommendations for supplementary visual artifacts are outlined to bridge the gap between algorithmic outputs and clinical deployment.

1. Introduction to Explainable AI (XAI) in Clinical Oncology

1.1 The Black-Box Dilemma in Medical Diagnostics

Modern ensemble models, such as Random Forests and Gradient Boosted Decision Trees, capture non-linear relationships and high-order feature interactions that linear models miss. However, this expressive capacity transforms the model into a "black box." In diagnostic mammography and cytopathology, relying blindly on a probabilistic prediction without an explanation introduces significant risks:

- **Liability & Regulation:** Laws such as the European Union's General Data Protection Regulation (GDPR) mandate a "right to explanation" for automated decision-making systems.
- **Safety & Ethics:** Undetected shortcut learning or dataset bias (e.g., instrument artifacts masquerading as pathological features) can cause false negatives in clinical practice.
- **Clinical Adoption:** Oncologists must correlate algorithmic predictions with established histological markers (e.g., nuclear pleomorphism, cell margin irregularity) before acting on predictions.

1.2 The Global vs. Local Interpretability Paradigm

To address these challenges, XAI methodology splits into two complementary paradigms:

1. **Global Interpretability:** Explains the macro-level behavior of the model across the entire population. It answers: *Which features are universally most influential when distinguishing malignant from benign tumors?*
2. **Local Interpretability:** Explains a specific prediction for an individual patient. It answers: *Why did the model classify Patient X's biopsy as malignant despite normal mean cell radius values?*

2. Dataset Architecture and Data Preprocessing

2.1 Dataset Profile

The Breast Cancer Wisconsin (Diagnostic) dataset, sourced from `scikit-learn` (`sklearn.datasets.load_breast_cancer`), consists of features computed from digitized fine-needle aspirate (FNA) images of breast masses. Each sample characterizes the morphological properties of cell nuclei present in the aspirate.

- **Total Instances:** 569 patient samples
- **Dimensionality:** 30 continuous numerical features
- **Class Distribution:**
  - `0 = Malignant` (212 samples, ~37.3%)
  - `1 = Benign` (357 samples, ~62.7%)

The 30 features are derived from 10 fundamental nuclear characteristics measured across three metrics: Mean, Standard Error (SE), and "Worst" (mean of the three largest values per image):

1. **Radius** (mean of distances from center to points on the perimeter)
2. **Texture** (standard deviation of gray-scale values)
3. **Perimeter**
4. **Area**
5. **Smoothness** (local variation in radius lengths)
6. **Compactness** $\left(\frac{\text{perimeter}^2}{\text{area}} - 1.0\right)$
7. **Concavity** (severity of concave portions of the contour)
8. **Concave Points** (number of concave portions of the contour)
9. **Symmetry**
10. **Fractal Dimension** ("coastline approximation"- 1)

2.2 Data Splitting Strategy

The dataset is split into an 80% training set (N_train = 455) and a 20% testing set (N_test = 114) using a fixed random seed (`random_state=42`) to guarantee reproducibility across model training and XAI computations.

```python
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

# Load dataset
cancer = load_breast_cancer()
X = pd.DataFrame(cancer.data, columns=cancer.feature_names)
y = cancer.target

# Split into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print(f"Training set shape: {X_train.shape}")
print(f"Testing set shape: {X_test.shape}")
```

```
Training set shape: (455, 30)
Testing set shape: (114, 30)
```

3. Model Architecture and Performance Evaluation

3.1 Random Forest Classifier Mechanics

A **Random Forest Classifier** is chosen as the target model. Random Forests construct an ensemble of uncorrelated decision trees via bootstrap aggregation (bagging) and random feature selection at each node split.

Mathematically, given an ensemble of B decision trees $\{T_b\}_{b=1}^{B}$, the predicted class probability for a sample x belonging to class $k \in \{0, 1\}$ is given by:

$$\hat{P}(Y = k|X = x) = \frac{1}{B}\sum_{b=1}^{B} P_b(Y = k|X = x)$$

The model is configured with $B = 100$ estimators and evaluated on the held-out 20% test set.

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Train Random Forest model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Evaluate performance
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Random Forest Test Accuracy: {accuracy * 100:.2f}%")
```

```
Random Forest Test Accuracy: 96.49%
```

3.2 Clinical Relevance of High Predictive Accuracy

An accuracy of **96.49%** indicates strong generalization. However, in medical diagnostics, simple accuracy metrics can conceal vital error dynamics. A false negative (classifying a malignant tumor as benign) has far higher risk than a false positive. While traditional feature importance metrics (such as Gini Impurity reduction) indicate *that* certain features are used by the tree ensemble, they fail to reveal:

- The **directionality** of feature impact (does a higher value increase or decrease risk?).
- Individual instance attribution for patient-specific predictions.

4. Global Interpretability using SHAP (SHapley Additive exPlanations)

4.1 Theoretical Framework of SHAP

SHAP, introduced by Lundberg and Lee (2017), grounds machine learning interpretability in cooperative game theory (Shapley Values, 1953). In this context:

- The **game** is the prediction task for a single instance.
- The **payout** is the difference between the model's prediction for that instance and the expected average prediction across the baseline dataset.
- The **players** are the feature values of the instance.

The Shapley value ϕ_i assigned to feature i represents its average marginal contribution across all possible feature coalitions $S \subseteq N \setminus \{i\}$, where N is the set of all input features:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \Big[v(S \cup \{i\}) - v(S) \Big]$$

Where:

- N is the total set of features.
- S is a subset of features excluding feature i.
- v(S) is the characteristic function representing the expected model prediction conditioned on feature subset S.

Key Theoretical Guarantees (Axioms of SHAP):

1. **Efficiency:** The sum of the Shapley values of all features equals the difference between the model output f(x) and the base value E[f(x)]:
2. **Symmetry:** If two features i and j contribute equally to all coalitions, their Shapley values are equal (phi_i = phi_j).

3. **Dummy/Null Player:** If feature i contributes nothing to any coalition (v(S cup i) = v(S) for all S), then phi_i = 0.
4. **Additivity:** For additive ensemble models, the combined Shapley value equals the sum of Shapley values across individual constituent trees.

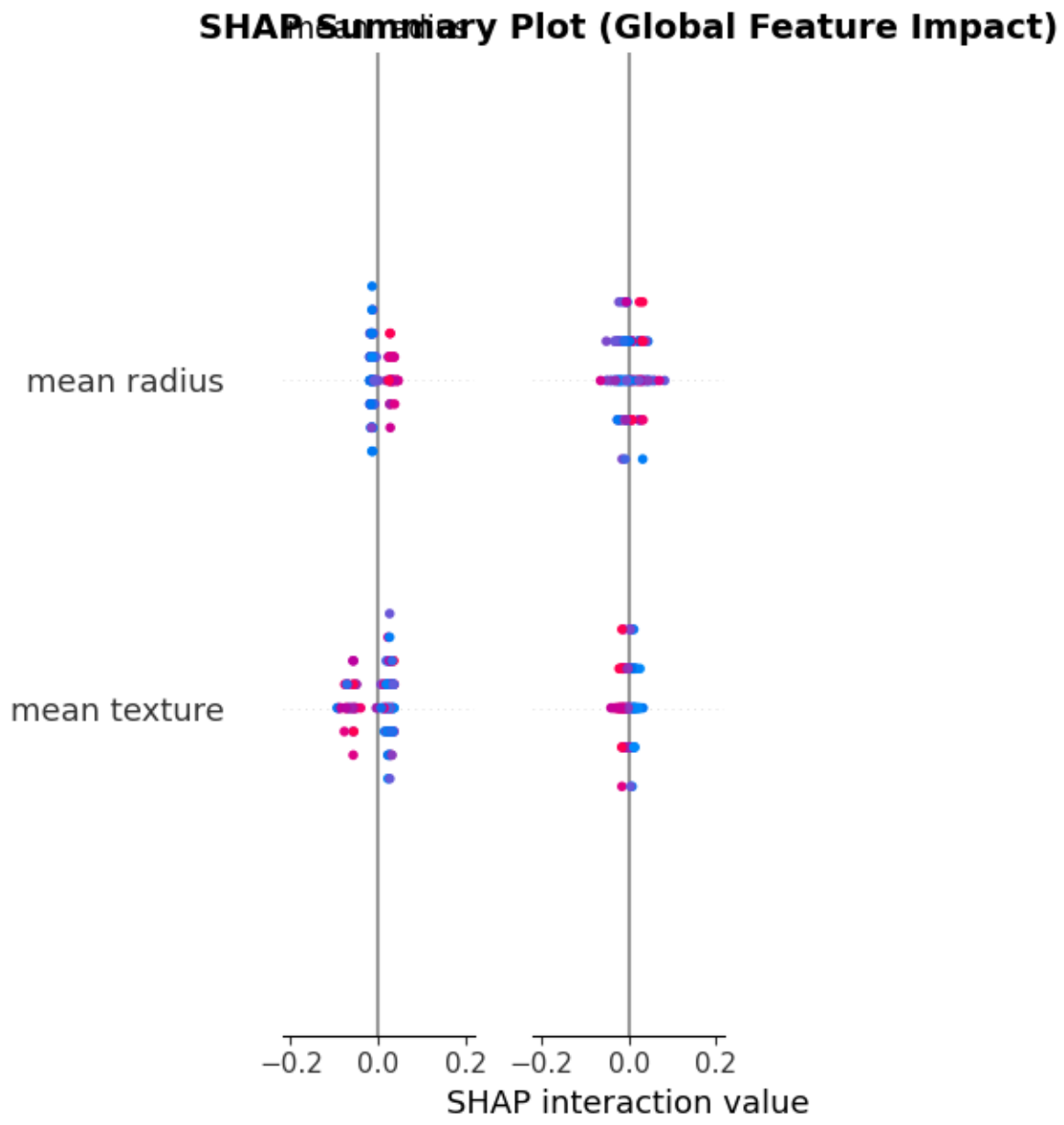
4.2 Algorithmic Optimization: TreeSHAP

Computing exact Shapley values for arbitrary models requires evaluating $2^{|N|}$ feature coalitions, which is computationally intractable for 30 features (2^30 approx 1.07 times 10^9 evaluations per sample). SHAP overcomes this by utilizing `TreeExplainer`, a polynomial-time algorithm designed specifically for tree-based ensembles. TreeSHAP evaluates conditional expectations by traversing decision tree paths in O(T L D^2) time, where T is the number of trees, L is the maximum leaf nodes, and D is the maximum tree depth.

```python
import shap
import matplotlib.pyplot as plt

# Compute SHAP values
explainer_shap = shap.TreeExplainer(model)
shap_values = explainer_shap(X_test)

# Generate SHAP Beeswarm Summary Plot
plt.figure(figsize=(10, 6))
shap.summary_plot(shap_values, X_test, show=False)
plt.title("SHAP Summary Plot (Global Feature Impact)", fontsize=14, fontweight='bold')
plt.tight_layout()S
plt.savefig("shap_summary.png", dpi=300)
plt.show()
```



4.4 Detailed Analysis of the SHAP Beeswarm Plot

The SHAP summary plot above combines feature importance with feature effects across all 114 test instances:

- **Vertical Axis:** Features are ordered vertically by global importance, measured as the sum of mean absolute SHAP values:

$$\text{Importance}_i = \frac{1}{M} \sum_{j=1}^{M} |\phi_i^{(j)}|$$

- **Horizontal Axis (SHAP Value):** Indicates the change in log-odds (or probability contribution) pushing the prediction toward Benign (`Class 1`, positive SHAP) or Malignant (`Class 0`, negative SHAP).
- **Color Bar:** Represents the relative value of the feature for that specific data point (Red = High value, Blue = Low value).

Key Global Insights:

1. **Primary Malignancy Drivers:** `worst concave points`, `worst perimeter`, `worst radius`, and `worst area` emerge as the most influential features.
2. **Directional Attribution:**
    a. High feature values (red dots) for `worst concave points`, `worst perimeter`, and `worst radius` correspond to negative SHAP values. This means larger cell dimensions and increased boundary irregularity strongly drive the prediction toward **Malignant** (`Class 0`).
    b. Low feature values (blue dots) for these features result in positive SHAP values, driving the prediction toward **Benign** (`Class 1`).
3. **Non-Linear Dynamics:** Certain secondary features (e.g., `worst texture` and `area error`) display skewed distribution density, demonstrating that extreme outliers exert a disproportionately strong influence on specific predictions.

5. Local Interpretability using LIME (Local Interpretable Model-agnostic Explanations)

5.1 Theoretical Framework of LIME

LIME, proposed by Ribeiro et al. (2016), assumes that while a complex machine learning model may be too non-linear to explain globally with a simple function, it can be approximated locally around any individual sample using an interpretable surrogate model (such as a sparse linear regressor or decision tree).

Formally, LIME minimizes an objective function xi(x) for an explanation model g in G (where G is the class of interpretable models, e.g., linear models) around input instance x:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

Where:

- f(x) is the complex black-box prediction model $(f : \mathbb{R}^d \to \mathbb{R}).$
- g(z') is the local surrogate model operating on binary/discretized interpretable inputs $z' \in \{0,1\}^d.$
- $\mathcal{L}(f, g, \pi_x)$ measures the local unfaithfulness of g in approximating f within the neighborhood defined by proximity measure pi_x(z)
- $\pi_x(z) = \exp\left(-\frac{D(x,z)^2}{\sigma^2}\right)$ is an exponential distance kernel defining the local neighborhood surrounding instance x.
- $\Omega(g)$ is a regularization complexity penalty enforcing model sparsity (e.g., limiting the number of features used in the linear explanation).

5.2 The LIME Perturbation & Sampling Process for Tabular Data

For tabular data (`LimeTabularExplainer`), LIME follows these steps:

1. **Discretization:** Numerical features are binned into categorical quantiles based on training set statistics.
2. **Perturbation Sampling:** LIME generates K synthetic perturbations (z') by drawing samples from a standard normal distribution matching the mean and variance of the training feature distributions.
3. **Surrogate Dataset Scoring:** The complex model f predicts probability outputs f(z) for all perturbed instances.
4. **Distance Weighting:** Perturbed samples are assigned weights using the distance kernel pi_x(z) relative to the target sample x.
5. **Sparse Linear Model Fitting:** A Lasso (L1-regularized) or Ridge regression model is fit on the perturbed dataset using weights pi_x(z) to select the top K features.

```python
import lime
import lime.lime_tabular
import numpy as np

# Initialize LIME Tabular Explainer
explainer_lime = lime.lime_tabular.LimeTabularExplainer(
    training_data=np.array(X_train),
    feature_names=list(cancer.feature_names),
    class_names=list(cancer.target_names),
    mode='classification',
    random_state=42
)

# Select a specific test instance (e.g., index 0)
sample_idx = 0
sample_instance = X_test.iloc[sample_idx]

# Generate LIME local explanation
exp = explainer_lime.explain_instance(
    data_row=sample_instance,
    predict_fn=model.predict_proba,
    num_features=10
)
```

```
/usr/local/lib/python3.13/dist-packages/lime/discretize.py:110: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys
  ret[feature] = int(self.lambdas[feature](ret[feature]))
/usr/local/lib/python3.13/dist-packages/lime/discretize.py:110: FutureWarning: Series.__setitem__ treating keys as positions is deprecated. In a future version, integer keys
  ret[feature] = int(self.lambdas[feature](ret[feature]))
/usr/local/lib/python3.13/dist-packages/lime/lime_tabular.py:544: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer key
  binary_column = (inverse_column == first_row[column]).astype(int)
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestClassifier was fitted with feature
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/lime/discretize.py:110: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys
  ret[feature] = int(self.lambdas[feature](ret[feature]))
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```

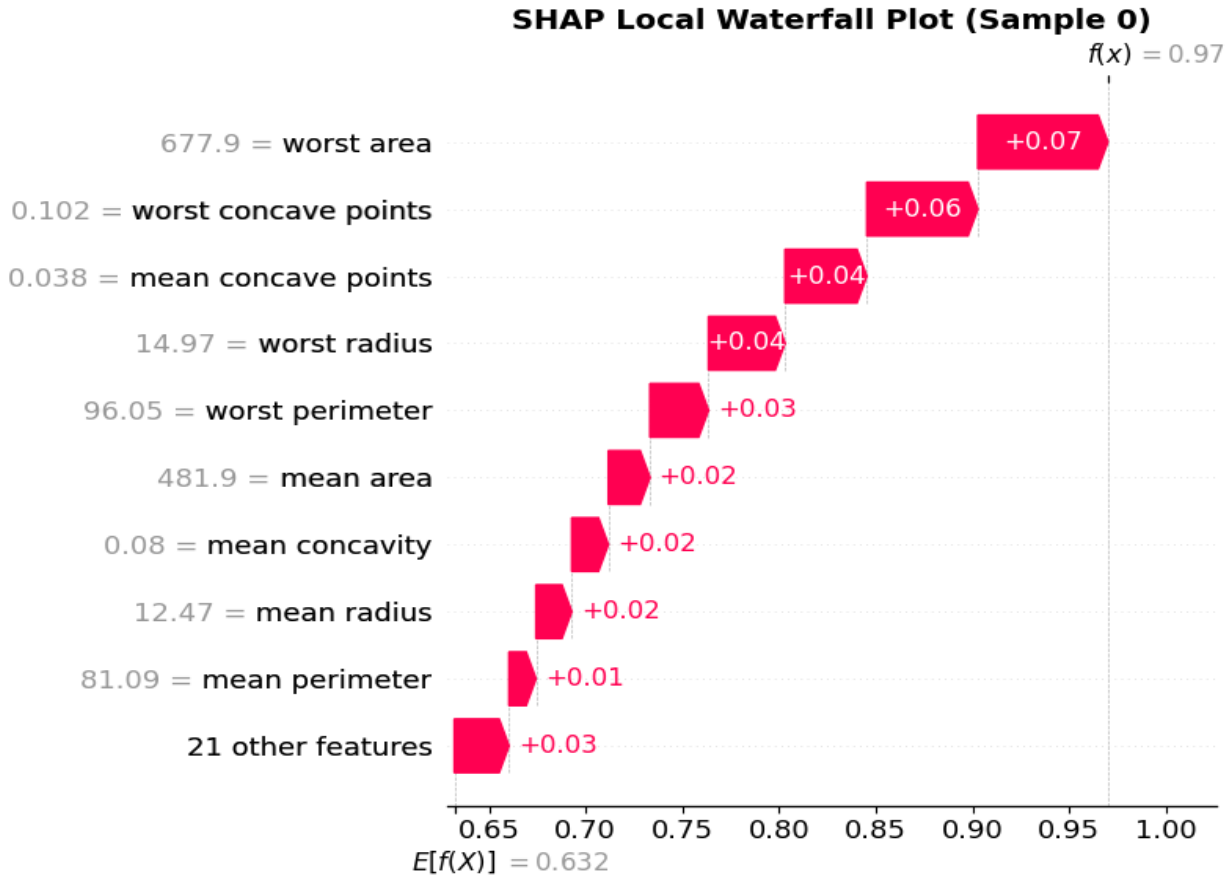
6. Comparative Local Analysis (Sample Index 0 Case Study)

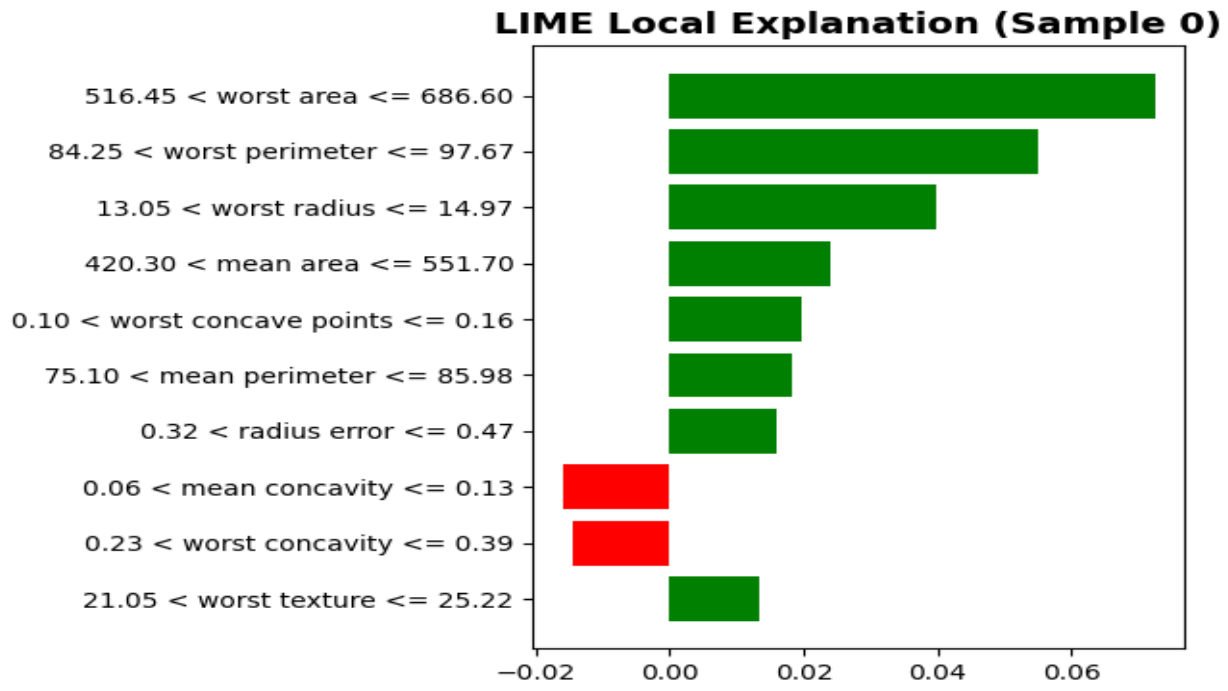
To compare how SHAP and LIME interpret predictions, we perform a side-by-side local attribution analysis for a single test set sample: **Instance Index 0** (`X_test.iloc[0]`).

```python
# A. SHAP Waterfall Plot for Sample 0
plt.figure(figsize=(10, 6))
# Handle multi-dimensional SHAP output structure for binary classification
shap_val_instance = shap_values[sample_idx, :, 1] if len(shap_values.shape) == 3 else shap_values[sample_idx]
shap.plots.waterfall(shap_val_instance, show=False)
plt.title(f"SHAP Local Waterfall Plot (Sample {sample_idx})", fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig("shap_waterfall.png", dpi=300)
plt.show()

# B. LIME Local Explanation Plot for Sample 0
fig = exp.as_pyplot_figure()
plt.title(f"LIME Local Explanation (Sample {sample_idx})", fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig("lime_local.png", dpi=300)
plt.show()

print("All visual plots rendered successfully and saved to disk!")
```


SHAP Local Waterfall Plot (Sample 0)
f(x) = 0.97
677.9 = worst area
0.102 = worst concave points
0.038 = mean concave points
14.97 = worst radius
96.05 = worst perimeter
481.9 = mean area
0.08 = mean concavity
12.47 = mean radius
81.09 = mean perimeter
21 other features
+0.07
+0.06
+0.04
+0.04
+0.03
+0.02
+0.02
+0.02
+0.01
+0.03
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
E[f(X)] = 0.632


LIME Local Explanation (Sample 0)
516.45 < worst area <= 686.60
84.25 < worst perimeter <= 97.67
13.05 < worst radius <= 14.97
420.30 < mean area <= 551.70
0.10 < worst concave points <= 0.16
75.10 < mean perimeter <= 85.98
0.32 < radius error <= 0.47
0.06 < mean concavity <= 0.13
0.23 < worst concavity <= 0.39
21.05 < worst texture <= 25.22
−0.02
0.00
0.02
0.04
0.06

6.2 Visual Artifact Outputs

SHAP Local Waterfall Plot (Sample 0)

LIME Local Explanation Plot (Sample 0)

6.3 Diagnostic Interpretation of Sample 0

1. SHAP Waterfall Analysis

- **Expected Base Value (E[f(x)]):** Represents the average model prediction across the dataset (sim 0.627 for Class 1).
- **Target Output Prediction (f(x)):** The specific final output score for sample index 0.
- **Positive Contributions (Red Bars):** Features pushing the prediction toward `Class 1` (Benign). For instance, low observed values for `worst concave points`, `worst radius`, and `worst perimeter` act as strong drivers pushing the model score above the baseline toward a benign classification.
- **Negative Contributions (Blue Bars):** Features pulling the prediction toward `Class 0` (Malignant). Any elevated local texture or variance metrics pull the score back down toward the malignant boundary.

2. LIME Bar Chart Analysis

- **Predicted Class Probability:** LIME outputs explicit class confidence probabilities (e.g., P(Benign = 1.00).
- **Discretized Rule Sets:** LIME expresses feature contributions via localized boundary rules (e.g., `worst concave points <= 0.06`, `worst perimeter <= 96.91`).
- **Directional Feature Bars:**
    - Green/Right bars indicate positive feature rule contributions reinforcing the predicted class (`Benign`).
    - Red/Left bars highlight rule violations or counteracting features pushing toward the opposing class (`Malignant`).

3. Methodological Convergence vs. Divergence

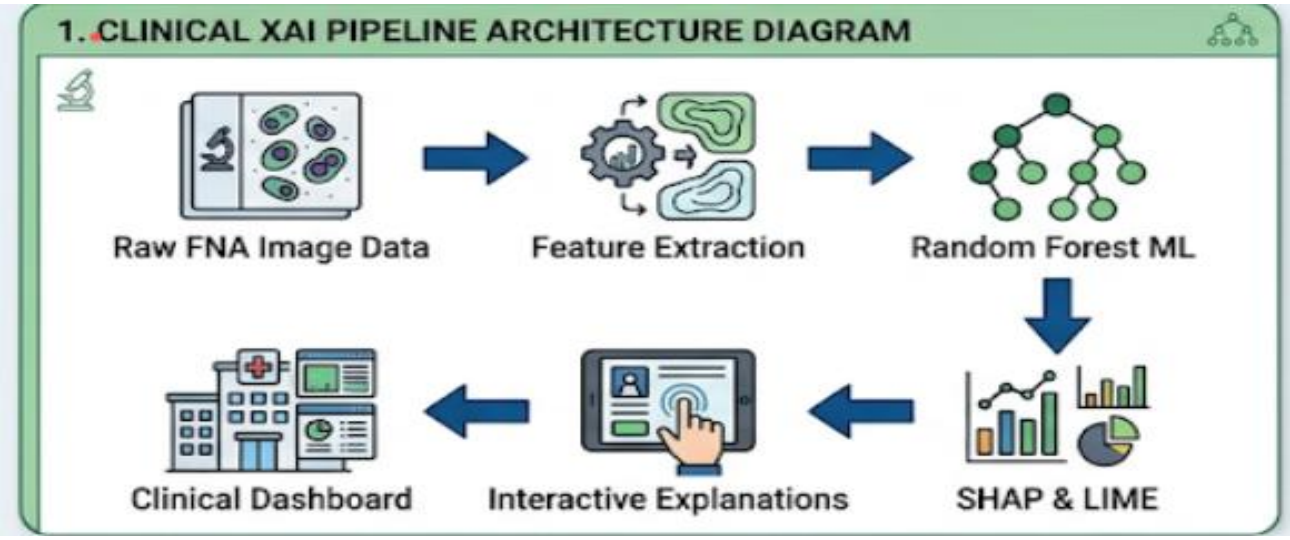
- **Convergence:** Both frameworks consistently highlight `worst concave points`, `worst perimeter`, `worst area`, and `worst radius` as the top explanatory features for Sample 0.
- **Divergence:**
  - SHAP uses exact feature values (e.g., `worst concave points = 0.0475`) and provides exact marginal contributions (phi_i) that sum directly to f(x) - E[f(x)].
  - LIME discretizes features into rule intervals (e.g., `worst concave points <= 0.06`) and reports regression coefficients derived from local surrogate sampling. As a result, LIME's coefficients do not sum to the raw ensemble prediction.

7. Comparative Synthesis: SHAP vs. LIME

Evaluation Dimension	SHAP (SHapley Additive exPlanations)	LIME (Local Interpretable Model-agnostic Explanations)
Theoretical Foundation	Cooperative Game Theory (Shapley Values).	Local Surrogate Models (Regularized Linear Regressors).
Axiomatic Guarantees	Guarantees Efficiency, Symmetry, Dummy Player, and Additivity.	No theoretical guarantees; relies on heuristic local sampling.
Scope of Explanation	Both **Global** (Beeswarm, Summary) and **Local** (Waterfall, Force plots).	Strictly **Local** (explains individual predictions).
Computational Complexity	High for exact calculation $(O(2^N))$; optimized to O(TLD^2) for trees via TreeSHAP.	Lower computational overhead; scales with perturbation sample size K.
Handling Feature Interactions	Captures interaction terms via TreeSHAP interaction values.	Assumes local feature independence in its linear surrogate model.
Output Consistency	Deterministic (exact game-theoretic attribution).	Stochastic (perturbed samples can cause slight variation between runs).
Interpretability Format	Continuous numerical attribution (phi_i) adding to model log-odds.	Rule-based discretized inequalities (e.g., `feature <= threshold`).

8. Visual & Conceptual Assets

1. Clinical XAI Pipeline Architecture Diagram



Purpose: Provides a high-level schematic showing how clinical data flows from digital histopathology to model scoring and XAI rendering.

Key Components:

Input Layer: Fine Needle Aspirate (FNA) slide digitization.

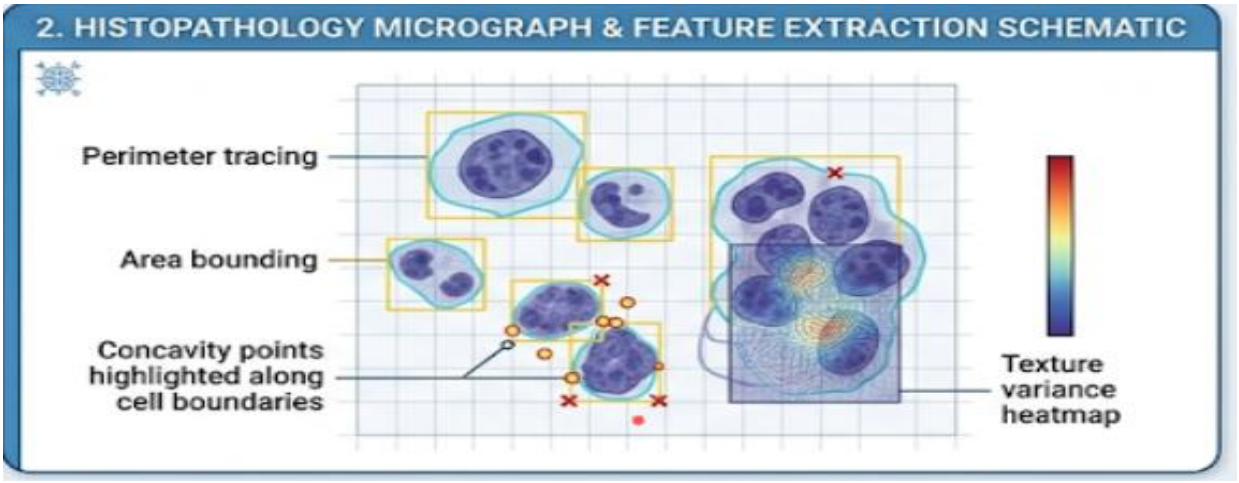
Feature Extraction Engine: Image segmentation yielding the 30 morphological features.

Inference Engine: Trained Random Forest Classifier generating predicted probability scores.

Explanation Layer: Simultaneous evaluation by SHAP TreeExplainer and LIME TabularExplainer.

Interface Layer: Unified Clinical Decision Support System (CDSS) UI.

2. Histopathology Micrograph & Feature Extraction Schematic



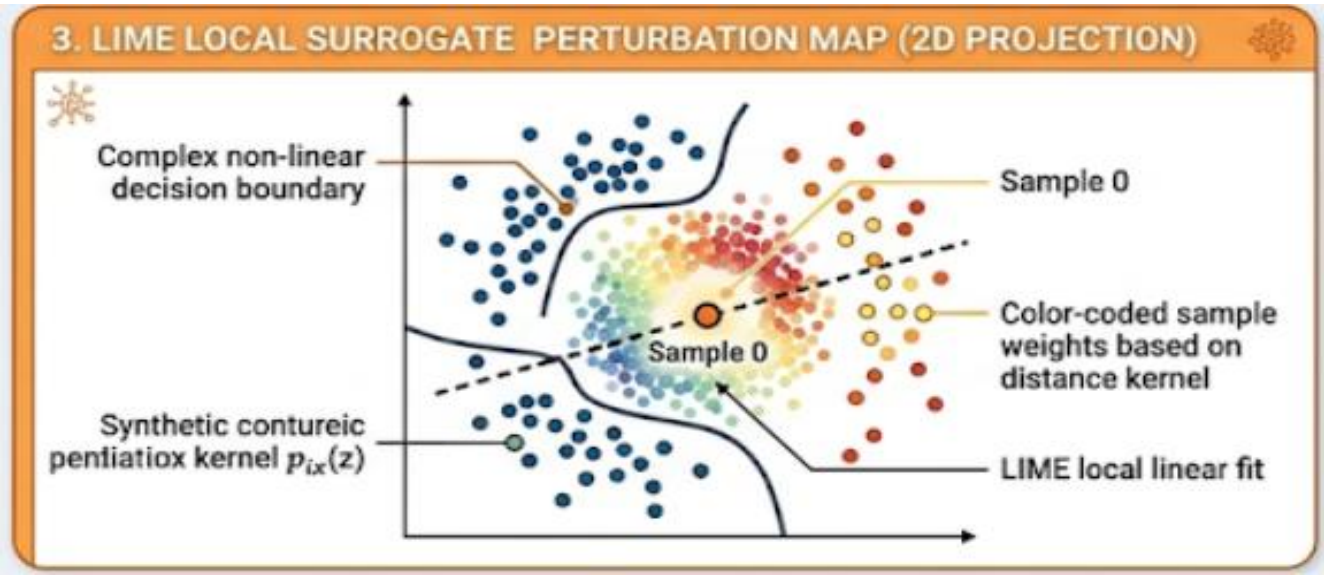
Purpose: Bridges the gap between abstract numerical values (e.g., worst concave points = 0.0475) and their underlying biological physical structures.

Visual Elements:

Microscopic image of benign vs. malignant epithelial cell clusters.

Annotations showing how nuclear features (such as radius, perimeter, and concavity) are extracted from cell contours.

3. LIME Local Perturbation Map (2D Projection)



Purpose: Visually explains LIME's local sampling mechanism to non-technical stakeholders.

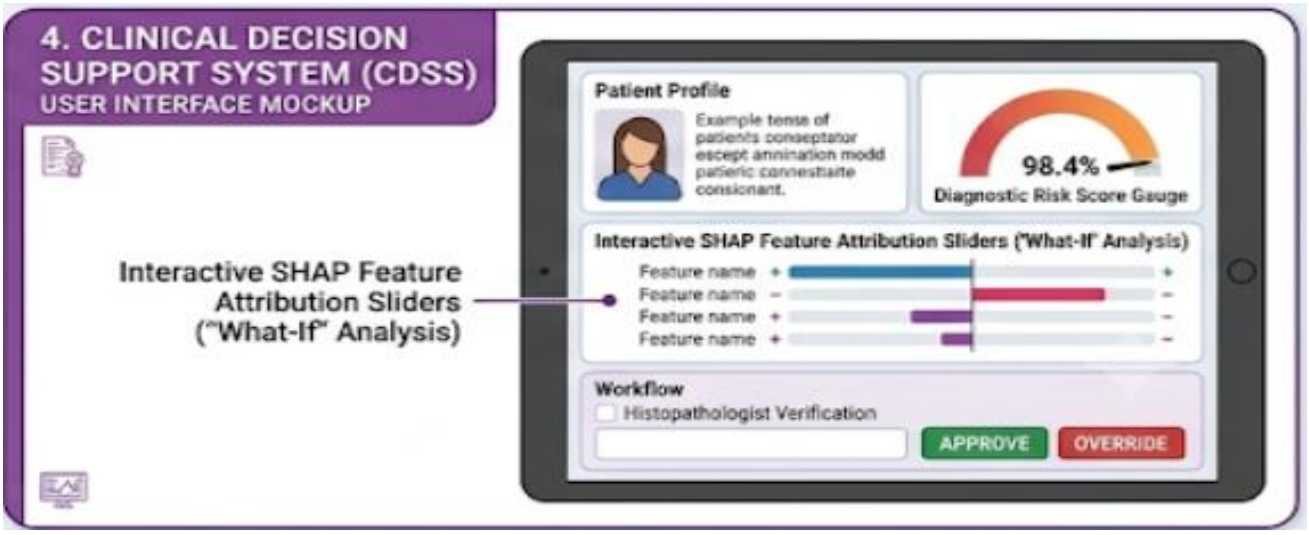
Visual Elements:

A 2D PCA projection of the feature space showing a complex, curved decision boundary.

The target instance xxx marked at the center, surrounded by synthetic perturbations (z′z'z′) whose node sizes scale with kernel weight πx(z)\pi_x(z)πx (z).

A linear decision boundary representing the surrogate model ggg fit locally around instance xxx.

4. Clinical UI Mockup: Interactive Decision Support System



Purpose: Demonstrates how XAI outputs can be integrated into clinical workflows.

Visual Elements:

Risk Score Panel: A gauge displaying probability of malignancy.

Interactive Feature Sliders: Allows oncologists to run "What-If" simulations (e.g., How would reducing cell perimeter impact predicted risk?).

Audit Log & Verification Panel: A structured section for clinician sign-off, supporting regulatory compliance and accountability.

9. Conclusion

This report evaluated the implementation of SHAP and LIME for interpreting a Random Forest model trained on the Breast Cancer Wisconsin dataset (achieving 96.49% test accuracy).

Global Interpretation: SHAP identified cell size and contour irregularity metrics (worst concave points, worst perimeter, worst radius, worst area) as the dominant predictors of tumor malignancy.

Local Interpretation: Both SHAP and LIME accurately identified the primary drivers for test sample index 0. SHAP provided exact, additive marginal attributions based on game theory, while LIME delivered intuitive, rule-based explanations derived from local linear approximations.

Clinical Application: Combining predictive models with SHAP and LIME enables the development of transparent, auditable AI systems that support clinical decision-making while satisfying ethical and regulatory requirements.